NAU · DEPARTMENT of MATHEMATICS and STATISTICS

Problem of the Week

#8

Fall 2026

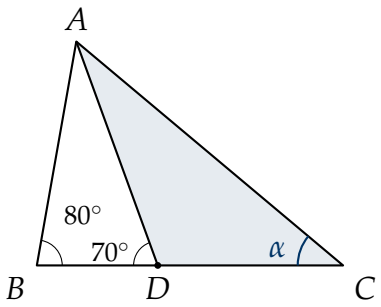
Due Monday, November 16 at 11:59 pm (Arizona time)

A Sum of Sides

In triangle ABC, the point D lies on side $\overline{BC}$. Suppose that

$$\angle ABC = 80^\circ, \qquad \angle BDA = 70^\circ, \qquad \text{and} \qquad AC = AB + BD.$$

Find the angle $\alpha = \angle BCA$, and prove that your answer is correct. (The figure is not needed for the solution.)



How to enter

Scan the code, or go to **ac.nau.edu/lms-apps/self-enroll/545026**, to join the POTW Canvas course. Then upload **one PDF** (typed, or neatly handwritten and scanned) to this week's assignment. Open to all NAU undergraduates.

Join on Canvas

The rules

- **Your own work only.** No AI tools (ChatGPT, Gemini, Claude, Copilot, Photomath, …), no internet, no calculators or software, and no help from other people.
- **Explain everything.** A bare answer earns little credit; a clear, complete argument earns full credit.
- Any participant may be asked to explain their solution in person.

Scored 0–4. Full rules, past problems, and model solutions:
naumathstat.github.io/problem-of-the-week
Questions: **ian.williams@nau.edu**